

Name: Manahil Habib

Section: 5-B

Reg no: 2023-BSE-042

Lab 7

Task 1

1. Print all environment variables:

```
manahil@ubuntu:~$ printenv

SHELL=/bin/bash
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29tZSAYMDAwMDAgMjAwMDAwMAA=
XDG_SEAT=seat0
PWD=/home/manahil
LOGNAME=manahil
XDG_SESSION_TYPE=tty
SYSTEMD_EXEC_PID=1394
HOME=/home/manahil
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:01:cd=40;33:01:or=40;31:01:mi=00:su=37;41:sg=30;43:ca=00:tw=30
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz
2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;3
31:*.r2=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;3
35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=0
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;3
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:.*~=00;90:.*#=00;90:*.bak=00;90:*.crdownload=00;90:
g-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
MEMORY_PRESSURE_WATCH=sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=8668bad1b1214e4eb9917f1af343042c
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=manahil
SHLVL=1
XDG_VTNR=1
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/ gnome:/usr/local/share:/usr/share:/var/lib/ snapd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
MAIL=/var/mail/manahil
_=/usr/bin/printenv
manahil@ubuntu:~$
```

2. Filter for SHELL, HOME and USER — run these greps together and capture one combined screenshot:

```
manahil@ubuntu:~$ 
manahil@ubuntu:~$ printenv | grep SHELL
SHELL=/bin/bash
manahil@ubuntu:~$ printenv | grep HOME
HOME=/home/manahil
manahil@ubuntu:~$ printenv | grep USER
USER=manahil
manahil@ubuntu:~$
```

Task 2

1. Define all DB_* variables (run the three exports one after another). Capture them in one screenshot showing the three export commands and their execution:

```
export DB_URL="postgres://db.example.local:5432/mydb"
```

```
USER=manahil
manahil@ubuntu:~$ export DB_URL="postgres://db.example.local:5432/mydb"
manahil@ubuntu:~$ export DB_USER="labuser"
manahil@ubuntu:~$ export DB_PASSWORD="labpass123"
manahil@ubuntu:~$ _
```

2. Echo the three variables (run the three echo commands together) and capture one screenshot showing their outputs:

```
manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ echo "$DB_USER"
labuser
manahil@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
manahil@ubuntu:~$ _
```

3. Show all DB_ variables with a single grep command (capture that output):

```
labpass123
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$
```

4. Close the bash session (e.g., exit) and reopen a new terminal. Verify the variables are gone by running the echo(s) and the grep together; capture both checks in one screenshot:

```
manahil@ubuntu:~$ echo "$DB_URL"
manahil@ubuntu:~$ printenv | grep '^DB_'
manahil@ubuntu:~$
```

Task 3

1. Open ~/.bashrc in an editor and append the three export lines. Capture the editor showing the three lines added (single screenshot):

```

# Lab 7 persistent DB variables
#
export DB_URL="postgres://db.example.local:5432/mydb"
#
export DB_USER="labuser"
#
export DB_PASSWORD="labpass123"
#
fi

```

2. Source ~/.bashrc and capture the source command in one screenshot together with the next verification commands (grouped): run source ~/.bashrc and then immediately run the three echoes and a single grep, capturing all of these in one screenshot:

```

manahil@ubuntu:~$ source ~/.bashrc
manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ echo "$DB_USER"
labuser
manahil@ubuntu:~$ echo "$DB_PASSWORD"
labpass123
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$

```

3. Close and reopen terminal. Verify persistence by running one echo and the grep together — capture both in one screenshot:

```

manahil@ubuntu:~$ echo "$DB_URL"
postgres://db.example.local:5432/mydb
manahil@ubuntu:~$ printenv | grep '^DB_'
DB_PASSWORD=labpass123
DB_USER=labuser
DB_URL=postgres://db.example.local:5432/mydb
manahil@ubuntu:~$

```

Task 4

1. View /etc/environment:

```

manahil@ubuntu:~$ sudo cat /etc/environment

PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
manahil@ubuntu:~$ _

```

2. Show current PATH (single print) — this is not a grep, but keep it as a dedicated screenshot:

```
manahil@ubuntu:~$ echo "$PATH"
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

3. Edit /etc/environment and add Class:

```
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<your_class_name>"
```

```
manahil@ubuntu:~$ sudo cat /etc/environment

PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
Class="CC-<your_class_name>"

manahil@ubuntu:~$
```

4. Re-login or open a new shell and show Class and PATH together (grouped prints):
run echo \$Class and echo \$PATH together and capture in a single screenshot:

```
manahil@ubuntu:~$ echo $Class
CC-<your_class_name>
manahil@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
manahil@ubuntu:~$
```

5. Create welcome script on Desktop and make it executable (capture the heredoc creation and chmod together in one screenshot if possible):

```
manahil@ubuntu:~$ cat > ~/Desktop/welcome <<'EOF'
>
> #!/bin/bash
>
> echo "Welcome to Cloud Computing $USER"
>
> EOF

manahil@ubuntu:~$
manahil@ubuntu:~$ chmod +x ~/Desktop/welcome
manahil@ubuntu:~$ ls -l ~/Desktop/welcome
-rwxrwxr-x 1 manahil manahil 55 Nov  6 04:32 /home/manahil/Desktop/welcome
manahil@ubuntu:~$
manahil@ubuntu:~$ _
```

6. Run the script from Desktop using ./welcome (dedicated screenshot):

```
manahil@ubuntu:~$ cd ~/Desktop
manahil@ubuntu:~/Desktop$ ./welcome
Welcome to Cloud Computing manahil
manahil@ubuntu:~/Desktop$
```

7. Add Desktop to PATH in ~/.bashrc (show the editor containing the PATH addition in one screenshot) and then source; capture the source and a single invocation of welcome (from home) together:

```
#
PATH=$PATH:~

~
~
~
~
~
~
~

".bashrc" 121L, 3789B written
manahil@ubuntu:~$ _
```

```
manahil@ubuntu:~$ source ~/.bashrc
manahil@ubuntu:~$ cd ~
manahil@ubuntu:~$ welcome
Welcome to Cloud Computing manahil
manahil@ubuntu:~$
```

Task 5

1. Enable ufw and show status (group both commands in one screenshot if you run them together):

```
manahil@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
manahil@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
3389/tcp ALLOW IN Anywhere
22/tcp ALLOW IN Anywhere
3389/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)

manahil@ubuntu:~$ _
```

2. Deny TCP port 22 and show status (run deny and status numbered together and capture in one screenshot):

```
manahil@ubuntu:~$ sudo ufw deny 22/tcp
Rule updated
Rule updated (v6)
manahil@ubuntu:~$ sudo ufw status numbered
Status: active
```

	To	Action	From
	--	-----	----
[1]	3389/tcp	ALLOW IN	Anywhere
[2]	22/tcp	DENY IN	Anywhere
[3]	3389/tcp (v6)	ALLOW IN	Anywhere (v6)
[4]	22/tcp (v6)	DENY IN	Anywhere (v6)

```
manahil@ubuntu:~$ _
```

3. From Windows host attempt to SSH (expected to fail) — capture the host-side SSH attempt in one screenshot:

```
ssh: connect to host 192.168.58.139 port 22: Connection timed out
PS C:\Users\HABIB TARIQ>
```

4. Allow SSH back and reload, then show status (group allow, reload, status in one screenshot if run together):

```
manahil@ubuntu:~$ sudo ufw allow 22/tcp
Rule updated
Rule updated (v6)
manahil@ubuntu:~$ sudo ufw reload
Firewall reloaded
manahil@ubuntu:~$ sudo ufw status
Status: active
```

To	Action	From
--	-----	----
3389/tcp	ALLOW	Anywhere
22/tcp	ALLOW	Anywhere
3389/tcp (v6)	ALLOW	Anywhere (v6)
22/tcp (v6)	ALLOW	Anywhere (v6)

```
manahil@ubuntu:~$ _
```

5. From Windows host attempt SSH again (should succeed) — capture successful login in one screenshot:

```
PS C:\Users\HABIB TARIQ> ssh manahil@192.168.58.139
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Nov  7 07:32:52 AM UTC 2025

System load:  0.13           Processes:            237
Usage of /:   94.2% of 9.75GB Users logged in:        1
Memory usage: 10%          IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

=> / is using 94.2% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov  6 08:44:48 2025 from 192.168.58.1
manahil@ubuntu:~$
```

Task 6

1. Generate ed25519 key pair (if needed) and show the generated files in one screenshot (run ssh-keygen and then list ~/.ssh):

```

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~ (main)
$ ssh-keygen -t ed25519 -f ~/.ssh/id_lab7 -C "lab_key"
Generating public/private ed25519 key pair.
Enter passphrase for "/c/Users/HABIB TARIQ/.ssh/id_lab7" (empty for no passphras
e):
Enter same passphrase again:
Your identification has been saved in /c/Users/HABIB TARIQ/.ssh/id_lab7
Your public key has been saved in /c/Users/HABIB TARIQ/.ssh/id_lab7.pub
The key fingerprint is:
SHA256:1EvhbYpnm7T9UkCiI2EoGYVm+i1ExbBJtJcLOHLW40s lab_key
The key's randomart image is:
+--[ED25519 256]--+
|oB*B.  .
|=*+=.  ..+.
|+*.o.  .oo
|.  +..o+  +.
|o.....S  .
|.o.  + =  .
|E.  +  ..
|
+-----[SHA256]-----+

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~ (main)
$

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~ (main)
$ ls -la ~/.ssh
total 29
drwxr-xr-x 1 HABIB TARIQ 197121  0 Nov  7 12:35 ./
drwxr-xr-x 1 HABIB TARIQ 197121  0 Nov  6 15:31 ../
-rw-r--r-- 1 HABIB TARIQ 197121 419 Oct  3 08:06 id_ed25519
-rw-r--r-- 1 HABIB TARIQ 197121 110 Oct  3 08:06 id_ed25519.pub
-rw-r--r-- 1 HABIB TARIQ 197121 399 Nov  7 12:35 id_lab7
-rw-r--r-- 1 HABIB TARIQ 197121  89 Nov  7 12:35 id_lab7.pub
-rw-r--r-- 1 HABIB TARIQ 197121 843 Oct 17 09:28 known_hosts
-rw-r--r-- 1 HABIB TARIQ 197121  97 Oct 17 09:28 known_hosts.old

```

2. Show the public key content (single screenshot):

type \$env:USERPROFILE\.ssh\id_ed25519.pub

or on Git Bash: cat ~/.ssh/id_ed25519.pub

```

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~ (main)
$ cat ~/.ssh/id_lab7.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGGkeJypQ0

```

3. Clear the known_hosts file content and verify it is empty (single screenshot):

Clear contents (PowerShell)

```

PS C:\Users\HABIB TARIQ> Clear-Content $env:USERPROFILE\.ssh\known_hosts
PS C:\Users\HABIB TARIQ> type $env:USERPROFILE\.ssh\known_hosts
PS C:\Users\HABIB TARIQ>

```

4. Connect to the Ubuntu server using the standard SSH command (this will prompt to accept the server host key because known_hosts is empty). Capture the connection prompt/accept step in one screenshot:

ssh username@<server_ip>

```
PS C:\Users\HABIB TARIQ> ssh manahil@192.168.58.139
The authenticity of host '192.168.58.139 (192.168.58.139)' can't be established.
ED25519 key fingerprint is SHA256:YrO7fyrK+VmpXQDjgLIQYvaMGVdry0WmrD0L8aXZSdE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.58.139' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Nov  7 07:42:52 AM UTC 2025

System load:  0.0          Processes:            238
Usage of /:   94.2% of 9.75GB Users logged in:      1
Memory usage: 11%         IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

=> / is using 94.2% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Fri Nov  7 07:32:53 2025 from 192.168.58.1
manahil@ubuntu:~$
```

5. After the successful connection, view the known_hosts file to show the server host key was added (single screenshot):

```
PS C:\Users\HABIB TARIQ> type $env:USERPROFILE\.ssh\known_hosts
192.168.58.139 ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIL8pGJwHHqC8/ZpIh+DjZ+MHkIpAef4igP50RAPMwJdu
192.168.58.139 ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQCiwonr8/0EUZ9xgdT2QtWvpKu93RD/6pRi44IvQPZ4LLjFwiiEwZxWJEzqsF6hB9YA
dfV0YagbSTx2vWYvCJL11FUyqG3SHAL+sT0LPTMOKt1akmMbDEzyy4Mc2dKlWEBHSYj0mECJ03A1P+00dLHKufBzcj+0p5GhMvTyBXH0KTZ4GPgdNeF0E0HP
IF+sr+Kz0G/sTtD5B3ZqkLpL4G9tiVv2jXiFGgWx0BqS8ojUZLvDGWYa9pYmqiljiAtgSSknJbNLZYoiS8GMycdcq+QteYv0STJTgp2bd5cTy1qkd0a+2mp
Bqt0/Lkd91MgkHEgix5zytjjiSfmBEjq3R7Lc0xbDLWPeRgSeKCa++uHxRc1l/uE10B/916UJvL7Ls1Vf1KmV2w0FoyW+jaq6yCS/L5IQ3i082VfAfbnj7FPA
AMzFME25pBtNsEwJmgh0LLpplmQzpIccvtY7GthXupkZ1DjPVHTdfFWp07Ss/KJ4LwM5NsT3982ugumdyBj5n0=
192.168.58.139 ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBBcZjBTgrwF65+pJ7J54z3GWTKYjCCNcK
Vsqd6R5sH57RZhxGa6GQT1odp3j+zfaTUzqa6wtKvMbaxVj2yv0X4A=
PS C:\Users\HABIB TARIQ>
```

B. On Ubuntu server — group related server-side commands:

1. Prepare the ~/.ssh directory and clear authorized_keys (this will create the directory if missing, set the correct directory permissions, and truncate the authorized_keys file). Capture this command sequence and its output in one screenshot:

```
manahil@ubuntu:~$ mkdir -p ~/.ssh
manahil@ubuntu:~$
manahil@ubuntu:~$ chmod 700 ~/.ssh
manahil@ubuntu:~$ > ~/.ssh/authorized_keys
manahil@ubuntu:~$ ls -ld ~/.ssh
drwx----- 2 manahil manahil 4096 Nov  7 07:33 /home/manahil/.ssh
manahil@ubuntu:~$ ls -l ~/.ssh/authorized_keys
-rw----- 1 manahil manahil 0 Nov  7 07:45 /home/manahil/.ssh/authorized_keys
manahil@ubuntu:~$ _
```

2. Append the public key, set file permissions, and show the resulting authorized_keys (capture commands and resulting file content in one screenshot):

```
manahil@ubuntu:~$ echo "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI...lab_key" > ~/.ssh/authorized_keys
manahil@ubuntu:~$
manahil@ubuntu:~$ chmod 600 ~/.ssh/authorized_keys
manahil@ubuntu:~$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI...lab_key
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGGkeJypQ0uGw+UZI
manahil@ubuntu:~$ _
```

3. From Windows host test passwordless login (capture successful login in one screenshot):

ssh username@<server_ip>

```
PS C:\Users\HABIB TARIQ> ssh manahil@192.168.58.139
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Nov  6 08:43:20 AM UTC 2025

System load:  0.01          Processes:            239
Usage of /:   94.6% of 9.75GB Users logged in:        1
Memory usage: 21%          IPv4 address for ens33: 192.168.58.1
Swap usage:   0%

=> / is using 94.6% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
10 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Nov  6 08:15:09 2025 from 192.168.58.1
manahil@ubuntu:~$
```

4. Also demonstrate explicit identity usage (single screenshot):

ssh -i ~/.ssh/id_ed25519 username@<server_ip>

```

PS C:\Users\HABIB TARIQ> ssh -vvv -i ~/.ssh/id_ed25519 manahil@192.168.58.139
OpenSSH_for_Windows_9.5p2, LibreSSL 3.8.2
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/config error:2
debug3: Failed to open file:C:/ProgramData/ssh/ssh_config error:2
debug2: resolve_canonicalize: hostname 192.168.58.139 is address
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts' -> 'C:\\Users\\HABIB TARIQ/.ssh/known_hosts'
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts2' -> 'C:\\Users\\HABIB TARIQ/.ssh/known_hosts2'
debug3: ssh_connect_direct: entering
debug1: Connecting to 192.168.58.139 [192.168.58.139] port 22.
debug1: Connection established.
debug1: identity file C:\\Users\\HABIB TARIQ/.ssh/id_ed25519 type 3
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert error:2
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert.pub error:2
debug3: failed to open file:C:/Users/HABIB TARIQ/.ssh/id_ed25519-cert error:2
debug1: identity file C:\\Users\\HABIB TARIQ/.ssh/id_ed25519-cert type -1
debug1: Local version string SSH-2.0-OpenSSH_for_Windows_9.5
debug1: Remote protocol version 2.0, remote software version OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
debug1: compat_banner: match: OpenSSH_9.6p1 Ubuntu-3ubuntu13.14 pat OpenSSH* compat 0x04000000
debug2: fd 3 setting O_NONBLOCK
debug1: Authenticating to 192.168.58.139:22 as 'manahil'
debug3: record_hostkey: found key type ED25519 in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:1
debug3: record_hostkey: found key type RSA in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:2
debug3: record_hostkey: found key type ECDSA in file C:\\Users\\HABIB TARIQ/.ssh/known_hosts:3
debug3: load_hostkeys_file: loaded 3 keys from 192.168.58.139
debug3: Failed to open file:C:/Users/HABIB TARIQ/.ssh/known_hosts2 error:2
debug1: load_hostkeys: fopen C:\\Users\\HABIB TARIQ/.ssh/known_hosts2: No such file or directory
debug3: Failed to open file:C:/ProgramData/ssh/ssh_known_hosts error:2
debug1: load_hostkeys: fopen _PROGRAMDATA_\\ssh/ssh_known_hosts: No such file or directory

```

System information as of Thu Nov 6 08:44:47 AM UTC 2025

System load:	0.08	Processes:	246
Usage of /:	94.6% of 9.75GB	Users logged in:	1
Memory usage:	22%	IPv4 address for ens33:	192.168.58.139
Swap usage:	0%		

=> / is using 94.6% of 9.75GB

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.

10 of these updates are standard security updates.

To see these additional updates run: apt list --upgradable

12 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Thu Nov 6 08:44:12 2025 from 192.168.58.1

manahil@ubuntu:~\$

Exam Evaluation Questions

Q1: Quick Environment Audit

- i. Run a single command to display environment variables and capture its output.

```

manahil@ubuntu:~$ printenv
SHELL=/bin/bash
DB_PASSWORD=labpass123
CREDENTIALS_DIRECTORY=/run/credentials/getty@tty1.service
MEMORY_PRESSURE_WRITE=c29t2SAyMDAwMDAgMjAwMDAwMAA=
XDG_SEAT=seat0
PWD=/home/manahil
LOGNAME=manahil
XDG_SESSION_TYPE=tty
SYSTEMD_EXEC_PID=1378
DB_USER=labuser
HOME=/home/manahil
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33:01;cd=40;33:01;or=40;31:01;mi=00:su=37;41:sg=30
x=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz
1;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.
2=01;31:*.t=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=0
31:*.r=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;
35:*.pnm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;3
:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;3
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.d
01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=0
36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:.*#=00;90:.*bak=00;90:.*cr
g-new=00;90:.*dpkg-old=00;90:.*dpkg-tmp=00;90:.*old=00;90:.*orig=00;90:.*part=00;90:.*rej=00;90:.*rpmnew=00;90:.*rpmorig=0
=00;90:.*ucf-dist=00;90:.*ucf-new=00;90:.*ucf-old=00;90:
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/system.slice/system-getty.slice/getty@tty1.service/memory.pressure
INVOCATION_ID=2c0a9b0e6d2d440f8484c21f5678b6c9
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=linux
LESSOPEN=| /usr/bin/lesspipe %s
USER=manahil
SHLVL=1
XDG_VTNR=1
DB_URL=postgres://db.example.local:5432/mydb
XDG_SESSION_ID=1
XDG_RUNTIME_DIR=/run/user/1000
XDG_DATA_DIRS=/usr/share/gnome:/usr/local/share:/usr/share:/var/lib/snapd/desktop
HUSHLOGIN=FALSE
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:/home/manahil/Desktop:/home/manahil
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
MAIL=/var/mail/manahil
Class=CC-BSE-Fall25
OLDPWD=/home/manahil
_=/usr/bin/printenv

```

- ii. In the same terminal session, run three filters (one per line) to show values for PATH, LANG, and PWD, then capture a single screenshot showing the three outputs together.

```

manahil@ubuntu:~$
manahil@ubuntu:~$ printenv PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:/home/manahil/Desktop:/home/manahil
manahil@ubuntu:~$ printenv LANG
en_US.UTF-8
manahil@ubuntu:~$ printenv PWD
/home/manahil
manahil@ubuntu:~$ _

```

Q2: Short-lived Student Info

- i. In one terminal, set three variables (STUDENT_NAME, STUDENT_ROLL_NUMBER, STUDENT_SEMESTER) using export — execute all three consecutively and capture them in one screenshot (show the commands executed).

```

manahil@ubuntu:~$ export STUDENT_NAME="Manahil"
manahil@ubuntu:~$ export STUDENT_ROLL_NUMBER="042"
manahil@ubuntu:~$ export STUDENT_SEMESTER="5"
manahil@ubuntu:~$

```

- ii. Still in the same session, print the three values with echo (grouped) and capture the outputs in one screenshot.

```
manahil@ubuntu:~$ echo $STUDENT_NAME
Manahil
manahil@ubuntu:~$ 
manahil@ubuntu:~$ echo $STUDENT_ROLL_NUMBER
042
manahil@ubuntu:~$ 
manahil@ubuntu:~$ echo $STUDENT_SEMESTER
5
manahil@ubuntu:~$
```

- iii. Use a single `printenv|grep` command to list any `STUDENT_` variables and capture the result.

```
manahil@ubuntu:~$ printenv | grep STUDENT_
STUDENT_NAME=Manahil
STUDENT_SEMESTER=5
STUDENT_ROLL_NUMBER=042
manahil@ubuntu:~$ 
manahil@ubuntu:~$ _
```

- iv. Exit that shell, open a fresh terminal, and show that the `STUDENT_` variables are not set (use `echo` and `printenv|grep` together) — capture in one screenshot.

```
manahil@ubuntu:~$ echo $STUDENT_NAME

manahil@ubuntu:~$ printenv | grep STUDENT_
manahil@ubuntu:~$ _
```

Q3: Make It Sticky (Persistence Check for Student Info)

- i. Edit `~/.bashrc` and append the three `STUDENT_*` exports. Capture a screenshot of the editor showing the new lines.

```
export STUDENT_NAME="Manahil"
export STUDENT_ROLL_NUMBER="042"
export STUDENT_SEMESTER="5"

-- INSERT --
```

- ii. Reload your shell config with a single command and then verify the three variables and show `printenv | grep '^STUDENT_'` — capture these verification outputs together in one screenshot.

```

manahil@ubuntu:~$ source ~/.bashrc
manahil@ubuntu:~$ echo $STUDENT_NAME
Manahil
manahil@ubuntu:~$ echo $STUDENT_ROLL_NUMBER
042
manahil@ubuntu:~$ echo $STUDENT_SEMESTER
5
manahil@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Manahil
STUDENT_SEMESTER=5
STUDENT_ROLL_NUMBER=042
manahil@ubuntu:~$ _

```

- iii. Close and re-open a terminal and demonstrate the STUDENT_NAME variable is available (echo and printenv grep together) — capture in one screenshot.

```

manahil@ubuntu:~$ echo $STUDENT_NAME
Manahil
manahil@ubuntu:~$
manahil@ubuntu:~$ printenv | grep '^STUDENT_'
STUDENT_NAME=Manahil
STUDENT_SEMESTER=5
STUDENT_ROLL_NUMBER=042
manahil@ubuntu:~$

```

Q4: Firewall Rules: Block and Restore Ping (ICMP)

- i. Enable ufw and capture the enable command and status together in one screenshot.

```

STUDENT_ROLL_NUMBER=042
manahil@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
manahil@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
3389/tcp ALLOW Anywhere
22/tcp ALLOW Anywhere
3389/tcp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
manahil@ubuntu:~$

```

- ii. Add a rule to block ping (ICMP echo) and show ufw status numbered in the same screenshot.

```

manahil@ubuntu:~$ sudo ufw deny proto icmp from any to any
ERROR: Unsupported protocol 'icmp'
manahil@ubuntu:~$ sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 3389/tcp ALLOW IN Anywhere
[ 2] 22/tcp ALLOW IN Anywhere
[ 3] 3389/tcp (v6) ALLOW IN Anywhere (v6)
[ 4] 22/tcp (v6) ALLOW IN Anywhere (v6)

manahil@ubuntu:~$ _

```

- iii. From your Windows host (or another client), attempt to ping the server while the rule is active and capture the blocked/failing ping in one screenshot.

```

PS C:\Users\HABIB TARIQ> ping 192.168.58.139

Pinging 192.168.58.139 with 32 bytes of data:
Request timed out.

```

- iv. Re-allow ping (ICMP) (or remove the deny rule) and capture the allow/reload/status sequence in one screenshot.

```

# ok icmp codes for INPUT
-A ufw-before-input -p icmp --icmp-type destination-unreachable -j ACCEPT
-A ufw-before-input -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-input -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-input -p icmp --icmp-type echo-request -j ACCEPT
# ok icmp code for FORWARD

```

```

manahil@ubuntu:~$ sudo ufw reload
Firewall reloaded
manahil@ubuntu:~$ sudo systemctl restart ufw
manahil@ubuntu:~$ _

```

- v. From the client, ping the server again and capture successful replies in one screenshot.

```

PS C:\Users\HABIB TARIQ> ping 192.168.58.139

Pinging 192.168.58.139 with 32 bytes of data:
Reply from 192.168.58.139: bytes=32 time=1ms TTL=64
Reply from 192.168.58.139: bytes=32 time=1ms TTL=64
Reply from 192.168.58.139: bytes=32 time=1ms TTL=64
Reply from 192.168.58.139: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.58.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\HABIB TARIQ> _

```
